CAMPUSAI

GREEN BUILDING (GB) – ONE-PAGE SAMPLE NOTES

1. Introduction

A green building is planned, designed, constructed and operated to reduce its environmental impact while providing a healthy and comfortable indoor environment for occupants.

2. Main Principles of Green Buildings

- **Energy efficiency:** Reduce energy use through insulation, efficient lighting, appliances and building design.
- **Water conservation:** Use water-efficient fixtures, detect leaks and collect rainwater where suitable.
- **Sustainable materials:** Choose durable, responsibly sourced or recycled materials with lower environmental impact.
- **Waste management:** Reduce construction waste and encourage reuse, segregation and recycling.
- **Indoor environmental quality:** Provide suitable ventilation, daylight, thermal comfort and low-emission materials.

3. Energy and Water-Saving Methods

- Use daylight and natural ventilation where the climate and design allow.
- Install LED lighting and energy-efficient equipment.
- Consider solar energy and other suitable renewable sources.
- Use rainwater harvesting and reuse treated greywater where permitted.
- Monitor consumption and maintain plumbing and electrical systems.

4. Benefits of Green Buildings

They can reduce energy and water consumption, lower operating costs over time, reduce waste and emissions, and improve occupant comfort and indoor air quality.

5. Example

A college building with shaded windows, LED lights, rooftop solar panels, rainwater harvesting, waste segregation and good natural ventilation uses several green-building practices.

6. Quick Revision Questions

1. Define a green building.
2. Explain any four energy-saving methods.
3. What is rainwater harvesting?
4. List the benefits of sustainable building materials.

Note: These are introductory sample notes. Refer to your course syllabus and class materials for complete coverage.